Testbench	A	A#	B	B#	Sum	OF_S	Diff	OF_D	LessThan
1	0110000110101000	25000	1011000111100000	-20000	5000	0	45000	1	0
2	1111110000011000	-1000	0000011111010000	2000	1000	0	-3000	0	1
3	0001111101000000	8000	0000001111101000	1000	9000	0	7000	0	0
4	1001111001011000	-25000	1011000111100000	-20000	-45000	1	-5000	0	1
5	0010011100010000	10000	1110110001111000	-5000	5000	0	15000	0	0
6	1001111001011000	-25000	0010011100010000	10000	-15000	0	-35000	1	1
7	0010011100010000	10000	0111010100110000	30000	40000	1	-20000	0	1
8	1111110000011000	-1000	1111100000110000	-2000	-3000	0	1000	0	0
range for 16 bits: -32768 to +32767									